SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer **ALL** questions in this section. Marks are indicated at the end of each question. Together they are worth 80% of the total marks for this examination.

Question 1 (20 marks – approximately 36 minutes)

24/7 Fitness Club is based in Hong Kong, operating 24 hours for seven days a week offering state-of-the-art equipment and facilities for group exercise exhilaration and personal training to fulfil individualised fitness aspirations to tone the body, build muscle, relieve tension, or get energised.

To expand the operations by offering more corporate training programs to those middle-income professionals working in the Central District, additional funding of HK$50 million is required. The company intends to raise the fund by issuing ten-year bonds with face value amounted to HK$1,000 for this purpose. Assume the required return on the bond issue would be 8%.

Required:

(a) Calculate the number of the bonds needed to be issued in order to raise the required funding of HK$50 million under each of the following scenarios:

(i) The bonds have a 5% coupon rate with semi-annual coupon payments.

(ii) The bonds make no coupon payments.

Round the bond prices to two decimal places and round up the number of bonds to the nearest integers.

(6 marks)

(b) Calculate the amount of repayment upon maturity of the bonds under each of the following scenarios:

(i) 5% coupon bonds mentioned in Question 1(a)(i).

(ii) Bonds without coupon payment mentioned in Question 1(a)(ii).

(4 marks)

(c) State the name of the bond without coupon payment mentioned in Question 1(a)(ii). Explain the reason for the different repayment amounts calculated in Questions 1(b)(i) and 1(b)(ii), and the advantages for issuing this bond from both the issuer's and from the bondholder's perspectives.

(6 marks)

(d) Explain TWO reasons why most companies prefer to use debt financing instead of equity financing.

(4 marks)

Question 2 **(20 marks – approximately 36 minutes)**

Your-car Manufacturing Company Limited ("YMCL") manufactures and exports electric automatic cars all over the world with its headquarter based in Tokyo. Some of the manufacturing parts are sourced in the United Kingdom ("UK"). Exports of the company's outputs specifically tailor-made for the UK market grows steadily which has been increasing to about 35% of YMCL's total sales turnover over the recent years. YMCL needs to convert the sales in British Pound to Japanese Yen regularly.

After the UK's decision in the 23 June 2016 public vote or referendum to leave the European Union ("EU"), volatility of the British Pound has been observed. The UK officially withdrew from the EU on 31 January 2020. During the transition period, major Union law is still applicable to the UK. YMCL exports goods to the UK utilising the free trade agreement between the EU and Japan. It has concerns over whether this framework can be maintained.

YMCL has a floating rate loan of JPY200 billion outstanding arising from the expansion of new manufacturing plants in other parts of Japan.

Required:

(a) Define and explain the purpose of the key risk indicator ("KRI"), and describe the process in developing a KRI program for the risk management.

(5 marks)

(b) Identify major financial risks faced by YMCL.

(4 marks)

(c) Explain how to mitigate risk arising from the foreign sales receipt generated from the export to the UK.

(3 marks)

(d) Explain how to mitigate risk arising from the outstanding loan of JPY200 billion.

(4 marks)

(e) List the FOUR financial risk management approaches.

(4 marks)

<u>Question 3</u> (20 marks – approximately 36 minutes)

Sweet Candy Company ("SCC") is a reputable wholesaling confectionery store chain providing the most common varieties of sugar confectionery and chocolates. SCC values efficient order management system because it ensures that all the sweets orders placed will be available at the door step as the delivery commitment to make the perfect accompaniment to all joyful occasions.

SCC has three product divisions and their financial results for 2020 were as follows:

In HK$' million	Cookies	Chocolates	Ice-creams
Sales turnover	105,000	38,000	15,000
Contribution margin	23,000	9,000	7,000
Net profit	7,000	3,500	2,900
Investment in total assets	36,000	12,000	18,000
Required rate of return	12.0%	18.0%	23.0%

SCC likes to maximise the total profitability, as well as to establish a performance improvement project to select the right key performance indicators ("KPIs") to assess and analyse its processes so as to evaluate success to achieve corporate goals.

<u>Required:</u>

(a) Define KPIs and describe key features which KPIs should possess for the performance evaluation.

(6 marks)

(b) Calculate the return on investment ("ROI") and residual income ("RI") for each division.

(6 marks)

(c) Analyse the performance of each division based on the answers calculated in Question 3(b), and explain which of the performance measures (i.e. ROI and RI) is more consistent with the corporate goal.

(5 marks)

(d) Apart from using financial measures, list THREE non-financial measures for the performance evaluation.

(3 marks)

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<u>Question 4</u> **(20 marks – approximately 36 minutes)**

Smart investment is a famous wealth management consultancy firm. After conducting financial analysis and risk assessment test, you are recommended to invest into a portfolio of equities with different economic sectors to achieve the maximum investment return with well-diversified risks.

The following is the information of the advised investment portfolio comprised of three stocks, A, B and C, in the financial industry, retailing business and natural resources respectively, without correlation with each other. You have invested 30% of your funding in Stocks A and B each, and 40% in Stock C.

		Rate of return if state occurs		
State of economy	Probability of state of economy	Stock A	Stock B	Stock C
Boom	0.6	15%	23%	-8%
Recession	0.4	4%	-19%	16%
	Beta	$\beta_A = 0.93$	$\beta_B = 1.86$	$\beta_C = 2.85$

<u>Required:</u>

(a) **Compute the expected returns for each stock. Calculate the portfolio returns under each state of economy and the expected return for the portfolio.**

Round % to two decimal places.

(4 marks)

(b) **Calculate the standard deviations for each stock and the portfolio.**

Round % to two decimal places.

(4 marks)

(c) **Compute the portfolio standard deviations using the weighted average of individual stock's standard deviation, and explain the rationale for the difference as compared with the answer computed in Question 4(b).**

(8 marks)

(d) **Identify and explain which stock has the highest total risk and which stock should be rewarded with the highest expected return.**

(4 marks)

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